LESSON 2

Fishing and Hunting

Objectives:



At the end of this lesson, you should be able to:

- identify the detrimental effects that human population caused to the environment in the context of fishing and hunting; and
- recommend possible human activities that can alleviate environmental concerns on fishing and hunting.



Introduction:

Hi, how are you? I hope you are doing well! Welcome to Lesson 2 of Module 1. In this lesson, we are going to explore the fishing and hunting activities of humans that affected ecological balance. For better understanding of these topics, there are series of activities that you need to accomplish that will guide you in your wider and deeper exploration. Are you ready? Let's begin!

Activity:



Unscramble the words in Column A, write your answer in Column B and determine which sector those words belong (<u>Fishing or Hunting</u>) and write your answer in Column C.

Column A	Column B	Column C
1. IHRFEIESS		
2. OAICGNPH		
3. SAACRCS		
4. WATRSL		
5. GOIIEVFSRHN		
6. IIOETCTNNX		

Analysis:

Fishing and hunting are common human activities. To what extent these activities start causing damages to the environment? Analyze the images on Figure 1.1 and Figure 1.2 below to help you expound your answer.



Figure 1.1 Overfishing



Figure 1.2 Poaching

_____

_____

_____

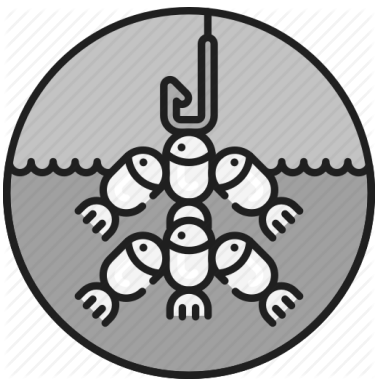
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Abstraction:

As you can see, fishing and hunting do not always cause harm to nature. However, human activities that go beyond consumption are the events that trigger environmental destruction. Now, let's discuss further other important details you should know about this topic.

Let's start with the definition.



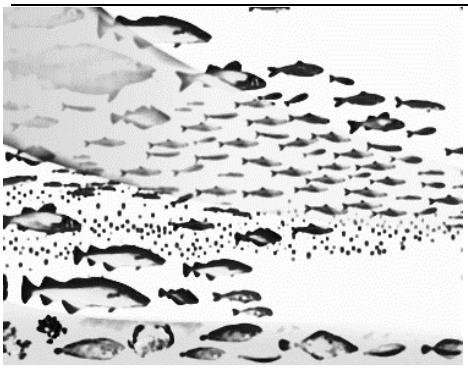
A **fishery** (plural: **fisheries**) is an organized effort by humans to catch fish or other aquatic species, an activity known as fishing.

The ecosystems that support fisheries, together with other economic activities, are subject to a **number of alterations** of significant relevance to their functioning and resilience and to the goods and services they can provide.

THE DETRIMENTAL EFFECTS OF FISHERIES

Fisheries impact target resources. They reduce their abundance, spawning potential and, population parameters (growth, maturation, etc.)

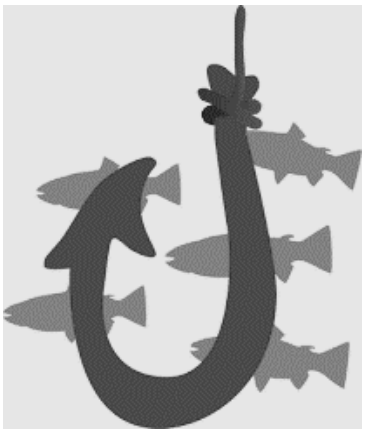




They modify age and size structure, sex ratio, genetics and species composition of the target resources, as well as of their associated and dependent species.

When poorly controlled, fisheries develop excessive fishing capacity, leading to overfishing, with major ecosystem, social and economic consequences. **Fishing** may also affect ecological processes at very large scale. The overall impact has been described as comparable, in aquatic systems, to that of agriculture on land in terms of the proportion of the system's primary productivity harvested by humans.

Figure 1.3 Stable Ecosystem



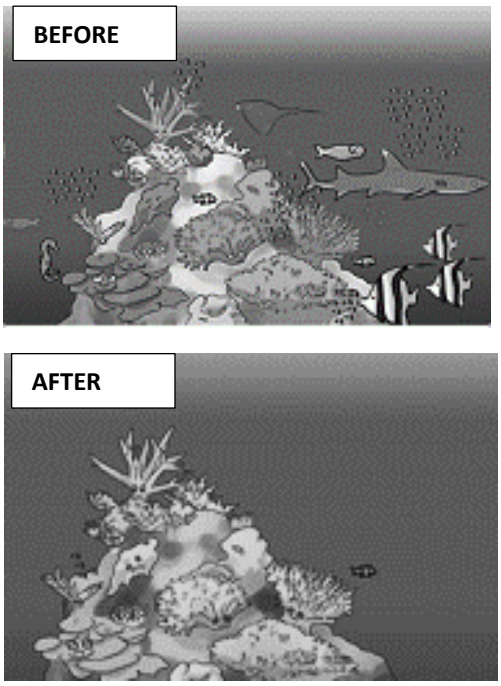
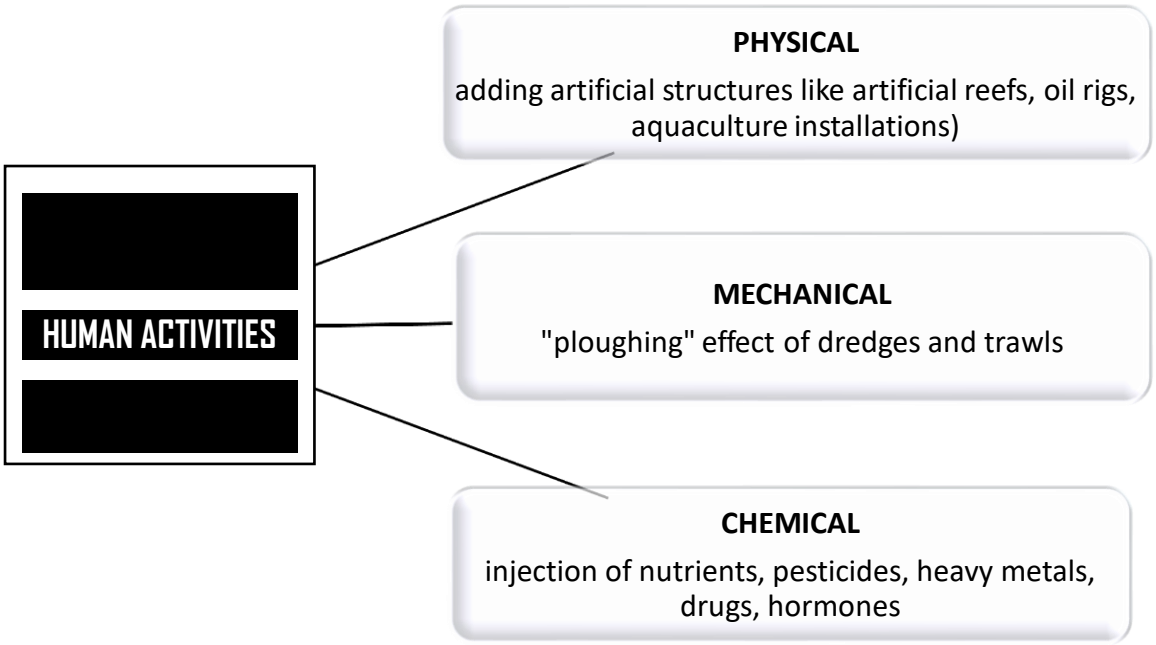


Figure 1.4 Stressed Ecosystem

OVERFISHING transforms an originally stable, mature and efficient ecosystem (see Figure 1.3) into one that is immature and stressed (see Figure 1.4). This happens in various ways. By targeting and reducing the abundance of high-value predators, fisheries deeply modify the trophic chain and the flows of biomass (and energy) across the ecosystem.

They can also alter habitats, most notably by destroying and disturbing bottom topography and the associated habitats such as (e.g. seagrass and algal beds, coral reefs) and benthic communities.



OVERFISHING may result in changes in productivity of resources.

DESTRUCTIVE FISHING TECHNIQUES

- using dynamite or cyanides or inadequate fishing practices (e.g. trawling in the wrong habitat)

POLLUTION FROM FISH PROCESSING PLANTS

- use of ozone-depleting refrigerants; dumping at sea of plastic debris that can entangle marine animals or be swallowed by turtles;

LACK OF SELECTIVITY

- affecting associated and dependent species, resulting in wasteful discarding practices, juvenile mortality, added threat to endangered species, etc.

IMPORTANT

Poorly-managed large-scale mariculture can damage coastal wetlands and nearshore ecosystems, often used as nurseries by key capture fishery resources, and contribute to ecosystem contamination with food residues, waste, antibiotics, hormones, diseases and alien species.



HUNTING TO EXTINCTION

Nowadays, most people hunt for sport, often leaving the carcass and taking the head, leaving the remains to decompose.



Because hunters are only allowed to pursue specific species, some environmentalists argue that hunting creates an imbalance in the natural elements of the environment (see Figure 1.5). Nature has a delicate balance and human hunting can have an impact on that natural balance.

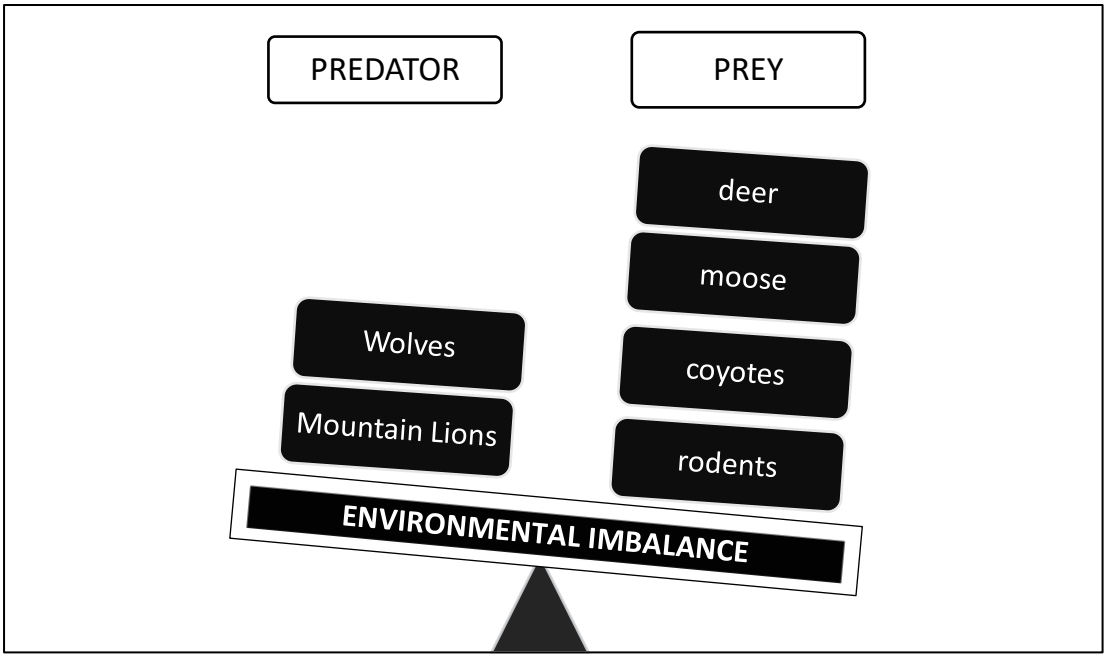


Figure 1.5 Diagram of an unbalanced ecosystem when predators are hunted to extinction

IMPORTANT

Opponents to hunting claim that animals have their own ways of population control and humans are not needed to aid that process.

DID YOU KNOW THAT?

The University of Michigan predicts over hunting will be the cause of extinction in about 25 percent of all animal extinctions during the 21st century. Whales and some African animals have become endangered due to hunting issues.

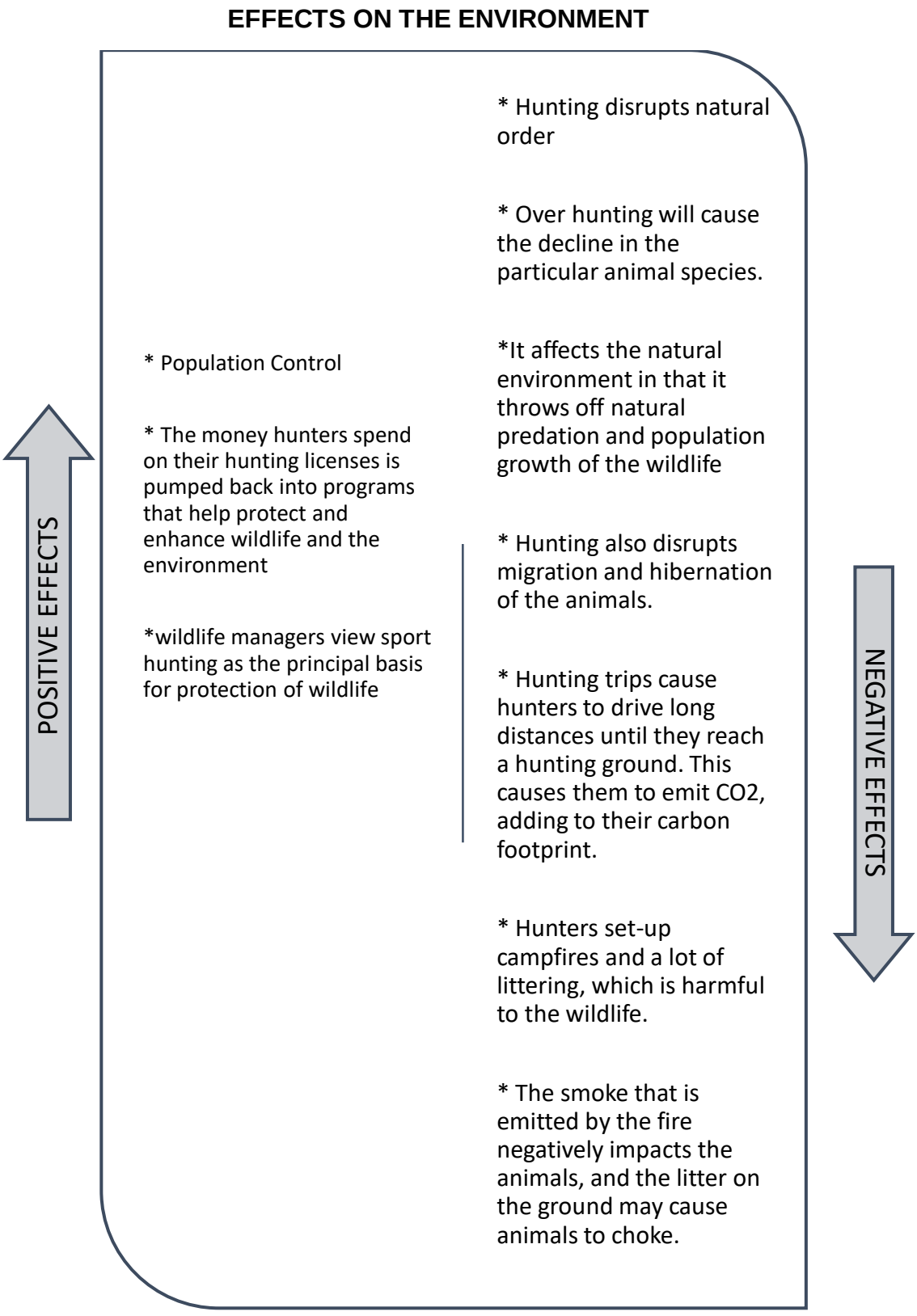


Figure 1.6 Effects of hunting to the environment

As shown in Figure 1.6, the negative effects of hunting to the environment outweigh the positive ones.

Application:



You have learned that if fishing and hunting are both properly regulated, negative effects to the environment can be avoided. Give at least (1) recommendation of what programs can be implemented to control fishing and hunting activities.

Fishing:

_____

_____

Hunting:

_____

_____

Assessment:



Match Column A with Column B to show the cause and effect relationship of the items below. Write only the letter of your answer.

COLUMN A

_____1. Hunting

_____2. Poorly-managed large-scale mariculture

_____3. Overfishing

_____4. Pursuing Specific Species of Predators

_____5. Long distances rides of hunters

COLUMN B

a. stable, mature and efficient ecosystem into one that is immature and stressed

b. More Emission of CO2

c. Damaged coastal wetlands and nearshore ecosystems

d. Extinction

e. Environmental Imbalance

Excellent! You just finished Lesson 2 of this Module! Should you have any questions or queries regarding this lesson, please do not hesitate to reach out to your instructors.

Are you ready to take your next lesson? Lesson 3 will discuss about Urban Industry. Good luck and enjoy reading.